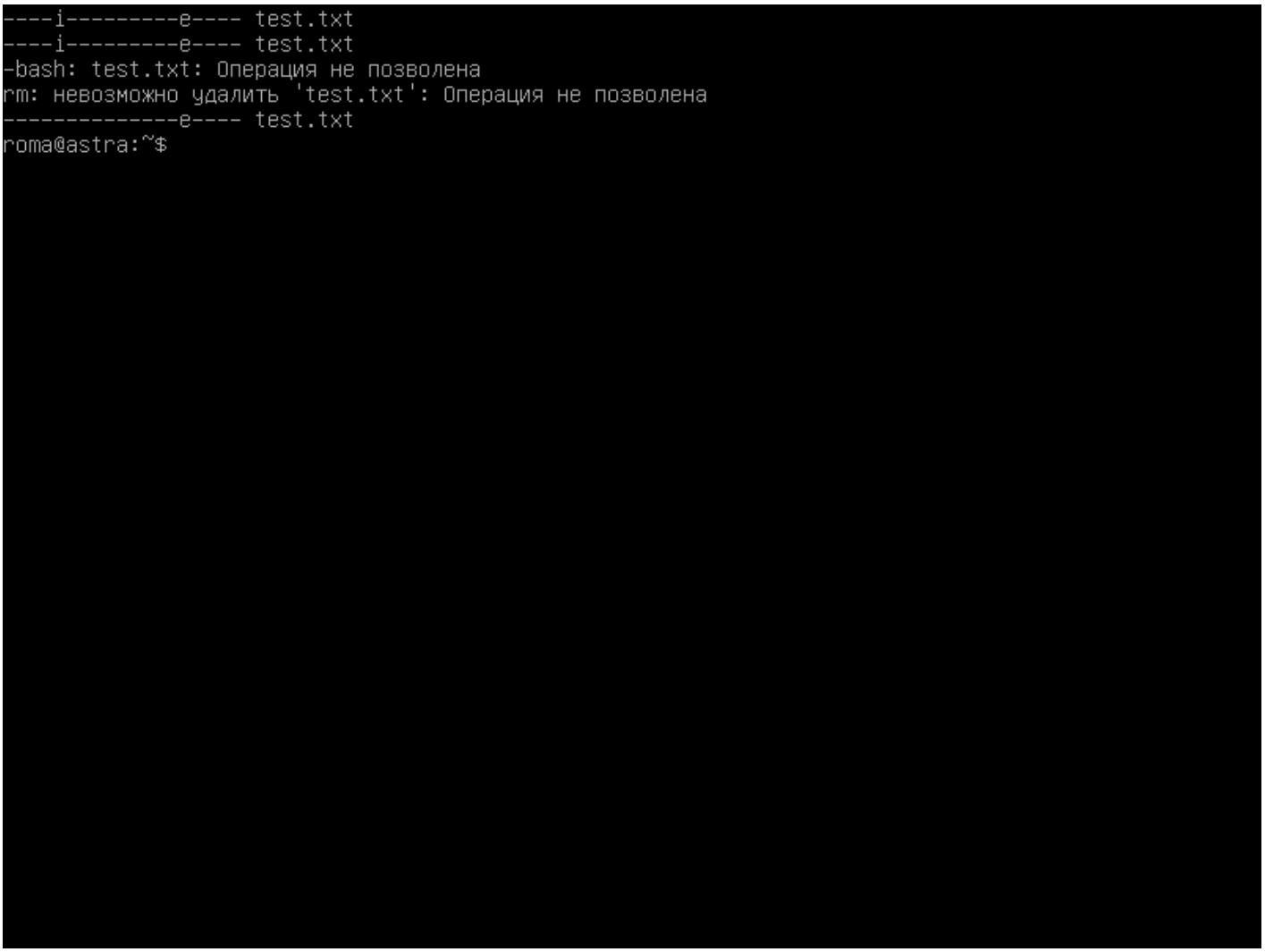
Лабораторная работа №3: Расширенные права доступа и аудит



```
getfacl: Removing leading '/' from absolute path names
# file: home/user1/acltest.txt
# owner: user1
# group: user1
user::rw-
user:user2:rw-
group::r--
mask::rw-
other::r--

created by user1
line from user2
user2 append
Для вас есть почта в /var/mail/roma
roma@astra:~$ _
```

```
*/2 * * * * date >> /home/user1/time.log
Пн ноя  9 19:00:01 MSK 2026
Пн ноя  9 19:02:01 MSK 2026
Пн ноя  9 19:04:01 MSK 2026
Пн ноя  9 19:06:01 MSK 2026
Пн ноя  9 19:08:01 MSK 2026
Пн ноя  9 19:10:01 MSK 2026
Пн ноя  9 19:12:01 MSK 2026
id: --context (-Z) работает только для ядер с поддержкой SELinux
roma@astra:~$ _
```

```
-w /etc/passwd -p rwxa -k passwd_watch
type=PATH msg=audit(09.11.2026 19:00:11.104:212) : item=1 name=/etc/ inode=262145 dev=fd:02 mode=dir
,755 ouid=root ogid=root rdev=00:00 nametype=PARENT cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_
frootid=0
type=PATH msg=audit(09.11.2026 19:00:11.104:212) : item=0 name=/etc/ inode=262145 dev=fd:02 mode=dir
,755 ouid=root ogid=root rdev=00:00 nametype=PARENT cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_
frootid=0
type=CWD msg=audit(09.11.2026 19:00:11.104:212) : cwd=/home/roma
type=SYSCALL msg=audit(09.11.2026 19:00:11.104:212) : arch=x86_64 syscall=rename success=yes exit=0
a0=0x7ffd02de6fa0 a1=0x6351f261bd60 a2=0x7ffd02de6f10 a3=0x73 items=5 ppid=1647 pid=1648 auid=roma u
id=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=tty1 ses=2 comm=u
seradd exe=/usr/sbin/useradd subj=? key=passwd_watch
roma@astra:~$ _
```

```
type=USER_LOGIN msg=audit(1794240084.464:468): pid=1726 uid=0 auid=4294967295 ses=4294967295 subj=?
msg='op=login acct=28696E76616C6964207573657229 exe="/usr/sbin/sshd" hostname=? addr=127.0.0.1 termi
nal=sshd res=failed'
----
time->Mon Nov  9 19:01:26 2026
type=USER_LOGIN msg=audit(1794240086.800:472): pid=1726 uid=0 auid=4294967295 ses=4294967295 subj=?
msg='op=login acct=28696E76616C6964207573657229 exe="/usr/sbin/sshd" hostname=? addr=127.0.0.1 termi
nal=sshd res=failed'
# date time acct host term exe success event
=========================================
1. 09.11.2026 19:01:22 nosuchuser 127.0.0.1 ssh /usr/sbin/sshd no 463
2. 09.11.2026 19:01:24 nosuchuser 127.0.0.1 ssh /usr/sbin/sshd no 467
3. 09.11.2026 19:01:26 nosuchuser 127.0.0.1 ssh /usr/sbin/sshd no 471
roma@astra:~$ _
```

```
  169  sudo cat /home/user1/time.log 2>&1 | tee /tmp/time1.txt; sudo ls -l /home/user1/time.log
  170  clear; ssh -o StrictHostKeyChecking=no -o UserKnownHostsFile=/dev/null nosuchuser@127.0.0.1
  171  sudo ausearch -m USER_LOGIN --success no 2>&1 | tail -12 | tee /tmp/aud_login2.txt
  172  sudo aureport -au 2>&1 | tail -12 | tee /tmp/aud_aureport2.txt
  173  sudo mount | grep -E ' / | /home ' | tee /tmp/mount_acl.txt
  174  sudo cat /home/user1/time.log; sudo ls -l /home/user1/time.log
  175  clear; date; sudo grep CRON /var/log/syslog | tail -8 | tee /tmp/cronlog.txt
  176  sudo cat /home/user1/time.log | tee /tmp/time2.txt
  177  clear; lsattr test.txt; sudo chattr +i test.txt; lsattr test.txt; echo 'change' >> test.txt;
rm -f test.txt; sudo chattr -i test.txt; lsattr test.txt
  178  clear; sudo getfacl /home/user1/acltest.txt; sudo -u user2 bash -c 'echo "user2 append" >> /h
ome/user1/acltest.txt'; sudo cat /home/user1/acltest.txt
  179  clear; sudo -u user1 -g crontab crontab -l; sudo cat /home/user1/time.log; id -Z
  180  clear; sudo auditctl -l; sudo ausearch -k passwd_watch -c useradd -i 2>&1 | tail -4
  181  clear; sudo ausearch -m USER_LOGIN --success no 2>&1 | tail -4; sudo aureport -au 2>&1 | tail
 -5
  182  cd /tmp
  183  touch acl_demo.txt
  184  sudo setfacl -m u:user2:rw acl_demo.txt
  185  getfacl acl_demo.txt
  186  sudo setfacl -x u:user2 acl_demo.txt
  187  getfacl acl_demo.txt
  188  sudo setfacl -m g:user2:r acl_demo.txt
  189  sudo setfacl -b acl_demo.txt
  190  getfacl acl_demo.txt
  191  cd ~
  192  echo "log line" > test2.txt
  193  sudo chattr +a test2.txt
  194  lsattr test2.txt
  195  echo "append line" >> test2.txt
  196  echo "overwrite" > test2.txt
  197  cat test2.txt
  198  sudo chattr -a test2.txt
  199  lsattr test2.txt
  200  cd ~; history -a; wc -l ~/.bash_history
  201  clear; history | tail -45
roma@astra:~$
```

```
1. 09.11.2026 19:01:22 nosuchuser 127.0.0.1 ssh /usr/sbin/sshd no 463
2. 09.11.2026 19:01:24 nosuchuser 127.0.0.1 ssh /usr/sbin/sshd no 467
3. 09.11.2026 19:01:26 nosuchuser 127.0.0.1 ssh /usr/sbin/sshd no 471
roma@astra:~$ cd /tmp
roma@astra:/tmp$ touch acl_demo.txt
roma@astra:/tmp$ sudo setfacl -m u:user2:rw acl_demo.txt
roma@astra:/tmp$ getfacl acl_demo.txt
# file: acl_demo.txt
# owner: roma
# group: roma
user::rw-
user:user2:rw-
group::r--
mask::rw-
other::r--

roma@astra:/tmp$ sudo setfacl -x u:user2 acl_demo.txt
roma@astra:/tmp$ getfacl acl_demo.txt
# file: acl_demo.txt
# owner: roma
# group: roma
user::rw-
group::r--
mask::r--
other::r--

roma@astra:/tmp$ sudo setfacl -m g:user2:r acl_demo.txt
roma@astra:/tmp$ sudo setfacl -b acl_demo.txt
roma@astra:/tmp$ getfacl acl_demo.txt
# file: acl_demo.txt
# owner: roma
# group: roma
user::rw-
group::r--
other::r--

roma@astra:/tmp$
```

```
other::r--

roma@astra:/tmp$ sudo setfacl -x u:user2 acl_demo.txt
roma@astra:/tmp$ getfacl acl_demo.txt
# file: acl_demo.txt
# owner: roma
# group: roma
user::rw-
group::r--
mask::r--
other::r--

roma@astra:/tmp$ sudo setfacl -m g:user2:r acl_demo.txt
roma@astra:/tmp$ sudo setfacl -b acl_demo.txt
roma@astra:/tmp$ getfacl acl_demo.txt
# file: acl_demo.txt
# owner: roma
# group: roma
user::rw-
group::r--
other::r--

roma@astra:/tmp$ cd ~
roma@astra:~$ echo "log line" > test2.txt
roma@astra:~$ sudo chattr +a test2.txt
roma@astra:~$ lsattr test2.txt
-----a--------e---- test2.txt
roma@astra:~$ echo "append line" >> test2.txt
roma@astra:~$ echo "overwrite" > test2.txt
-bash: test2.txt: Операция не позволена
roma@astra:~$ cat test2.txt
log line
append line
roma@astra:~$ sudo chattr -a test2.txt
roma@astra:~$ lsattr test2.txt
--------------e---- test2.txt
roma@astra:~$
```